

MASTER TECHNICAL REPORT

MULTI-TARGET ADVERSARIAL EVALUATION · STAGE 3

Red Lab fires adversarial and malformed inputs at a live text-classification endpoint and produces an evidence-backed robustness report, comparing an unhardened configuration against a hardened one wherever a hardened counterpart exists.

It is for developers and maintainers deciding whether application-layer hardening measurably changes how a text classifier's HTTP endpoint behaves under malformed, boundary and meaning-preserving adversarial input.

A developer adapts it to their own compatible text classifier by registering a new target in endpoint/targets.json (model id, revision, labels) and pointing the target-aware runner at it; the attack suites, analysis and this report generator do not need to change.

After judge feedback that the toolkit's Technical Report explained only the historical single-model emotion benchmark, V6.3 adds this master report: it covers all five approved evaluation identities (emotion and sentiment, core and OCES, plus the CI release gate), explains the target-aware architecture that made a second real target possible, and separates real from historical evidence explicitly.

This report demonstrates that specific, named adversarial and malformed inputs produced specific, reproducible, hash-verified behavior differences on these exact pinned model revisions at these exact commits. It does not prove general model safety, does not certify any model beyond the ones evaluated, and a persisting model-level finding after hardening is not evidence the hardening failed -- the four frozen defenses are application-layer only.

How the toolkit is built

Public application (web/)

FastAPI app serving Home, Lab and this report; orchestrates at most one live run at a time via `run_all.run_experiment`.

Registered targets (endpoint/targets.json)

A static, model-free registry of models, tasks, targets and the five evaluation identities. Loading it never imports torch/transformers/an endpoint app.

Attack generation and frozen suites (attacks/, baseline/)

Pre-registered oracle and validity-tier metadata per case, built before any result exists; a sidecar manifest is the only channel analysis reads from it.

Target-aware runner (runner/)

Sends the suite to one live endpoint, records exact request bytes/hashes, and (`runner.gate`) enforces a frozen CI policy against a candidate target.

Analysis and remediation (analysis/)

Scores coverage, drift and findings from recorded results plus the attack manifest; never invents pass/fail criteria analysis was not given.

Reports (report/)

Schema v2/v3 validated rendering, including this master report generator, which never re-scores evidence -- only renders it.

CI release gate (.github/workflows/release-gate.yml)

Runs `runner.gate` against the frozen policy on every PR into stage3/integration; deployment depends on the gate passing.

Emotion V1 (`emotion_v1`, unhardened) and Emotion V2 (`emotion_v2`, hardened) are two configurations of the same model and form the only paired comparison in this report. Sentiment V1 (`sentiment_v1`) is a single target with deliberately no V2: `sentiment.core` and `sentiment.oces` report comparison: null and carry no before/after claim. "Analysis schema v3" is the target-aware JSON data format every evaluation's `analysis.json` is written in -- it is a document shape, not a network endpoint or service.

The five approved evaluation identities

EVALUATION ID	KIND	TARGET ID(S)	SUITE	MODEL IDENTITY	PURPOSE
emotion.core	paired	emotion_v1, emotion_v2	core	j-hartmann/ emotion- english- distilroberta- base @ 0e1cd914e3d4 ...	Original V1/V2 emotion experiment; measures what application- layer hardening changes.
sentiment.core	single	sentiment_v1	core	distilbert/ distilbert-base- uncased- finetuned- sst-2-english @ 714eb0fa89d2 ...	Single-target sentiment robustness baseline; no hardened counterpart exists.
emotion.oces	paired	emotion_v1, emotion_v2	oces	j-hartmann/ emotion- english- distilroberta- base @ 0e1cd914e3d4 ...	Additional paired paraphrase/ distractor evaluation authored after the defenses were frozen.
sentiment.oces	single	sentiment_v1	oces	distilbert/ distilbert-base- uncased- finetuned- sst-2-english @ 714eb0fa89d2 ...	Additional single-target sentiment paraphrase/ distractor evaluation.
ci.emotion	single	emotion_v2	core	j-hartmann/ emotion- english- distilroberta- base @ 0e1cd914e3d4 ...	Automated release gate: emotion_v2 (candidate) against a frozen case selection and policy.

Emotion Core Results (historical, verified)

The original, previously verified benchmark: the same emotion model exposed as an unhardened endpoint (V1) and a hardened endpoint (V2), each sent the identical 1928-case suite. This section presents the existing verified figures; it is not re-run or re-scored here.

emotion_v1 — Unhardened			emotion_v2 — Hardened		
1370	218	29	1373	112	14
eligible flip comparisons	qualifying prediction flips	finding groups	eligible flip comparisons	qualifying prediction flips	finding groups

Same underlying model — V1 and V2 run the identical j-hartmann/emotion-english-distilroberta-base classifier; only the endpoint configuration differs.

Common eligible denominator — 1928 matched recorded cases form the common eligible base for the V1→V2 comparison (0 V1-only, 0 V2-only, 0 unavailable).

Findings resolved after hardening: 15 · **Findings remaining on V2:** 14

Resolved on V2: every malformed/boundary/over-limit-truncation unhandled-5xx finding present on V1 (oversized payloads, length_plus_one, over-limit truncation signals) is absent from V2's finding register -- the four frozen defenses (length limit, strict typing, scoped exception handling, Unicode/whitespace normalization) removed that entire class.

Persisting on V2: encoding and perturbation prediction-flip findings (homoglyphs, backspace deletion, typos, whitespace) appear on both V1 and V2 at similar rates, because the frozen defenses are application-layer only and were never expected to change the model's own sensitivity to meaning-preserving text transformations.

This is the historical, previously verified V5 benchmark. It is not re-run or re-scored here. Full finding registers, per-case evidence and the rendered historical report are archived at [/verified-full/](#) (see also [Historical Evidence](#)).

Sentiment Core Results

Single-target evaluation. sentiment_v1 is the only sentiment target; there is no sentiment_v2. No hardening comparison is available or implied, and no improvement percentage is reported.

RECORDED COVERAGE

1931 / 1931

100.00% of planned cases recorded

QUALIFYING PREDICTION FLIPS

6.17%

99 / 1604 eligible prediction pairs

Model distilbert/distilbert-base-uncased-finetuned-sst-2-english@714eb0fa89d2f80546fda750413ed43d93601a13, labels NEGATIVE, POSITIVE. 18 finding groups recorded. Operational: 85 unhandled 5xx, 1 timeouts, 1 invalid input accepted.

Medium · encoding/homoglyph · prediction_flip

ENCODING/HOMOGLYPH: PREDICTION FLIP

21 case(s) in encoding/homoglyph produced prediction flip -- the top label changed under a meaning-preserving transformation while the clean prediction was confident (at or above 0.6). Severity is scored from the single highest-scoring instance; the 21 affected case(s) indicate reproducibility, not impact.

Observed evidence

21 of 21 cited case row(s) were available; recorded outcomes: HTTP 200: 21; post-request health-check failures: 0.

Why it matters

The top label changed under a cross-script confusable substitution. Unicode compatibility normalisation does not fold these: confusables are a security mapping (UTS #39), separate from the compatibility mapping NFKC applies.

Recommended action

Mixed-script confusables need a confusables-aware input check rather than more normalisation. The model-level sensitivity remains underneath, so this reduces exposure to this family without making the classifier robust to it.

How to verify the fix

python -m runner.run --target-id sentiment_v1 --case-set results/v63_real_sentiment_core_bf0c268072bf48dd/runs/sentiment_core/verification_case_set.json --out results/verify_sentiment_v1

the same homoglyph case ids are re-run and the substituted characters are folded or the request is rejected

+ Evidence cases · 21

sst2-val-121.encoding.homoglyph.middle.d3; sst2-val-121.encoding.homoglyph.middle.d5; sst2-val-13.encoding.homoglyph.leading.d3; sst2-val-13.encoding.homoglyph.leading.d5; sst2-val-13.encoding.homoglyph.middle.d1; sst2-val-13.encoding.homoglyph.middle.d3; sst2-val-13.encoding.homoglyph.middle.d5; sst2-val-201.encoding.homoglyph.trailing.d5; sst2-val-341.encoding.homoglyph.leading.d3; sst2-val-341.encoding.homoglyph.leading.d5; sst2-val-35.encoding.homoglyph.middle.d1; sst2-val-35.encoding.homoglyph.middle.d3; sst2-val-35.encoding.homoglyph.middle.d5; sst2-val-588.encoding.homoglyph.middle.d3; sst2-val-588.encoding.homoglyph.middle.d5; sst2-val-588.encoding.homoglyph.trailing.d1; sst2-val-59.encoding.homoglyph.leading.d3; sst2-val-59.encoding.homoglyph.leading.d5; sst2-val-59.encoding.homoglyph.middle.d1; sst2-val-672.encoding.homoglyph.middle.d3; sst2-val-672.encoding.homoglyph.middle.d5;

TEAM CHECKMATE / ROBUSTNESS REPORT

5 / 19

OCES — Out-of-Coverage Evaluation Suite

OCES stands for the Out-of-Coverage Evaluation Suite: a small, separately frozen set of paraphrase and distractor cases, additional to the main core suite, exercised against the same live endpoints.

OCES was authored by this team after the evaluated defenses were already frozen, and the authors knew what those defenses were. It is not an external or blind holdout and must never be described as independent, design-independent, third-party or universally representative -- it is a separately frozen additional evaluation, nothing more.

Validity tiers (both targets combined)

0 GOLD · 50 SILVER · 32 REVIEW (82 cases total)

Case composition

42 emotion variants · 40 sentiment variants; exactly the approved `oces.paraphrase` and `oces.distractor` families.

emotion.oces (paired: `emotion_v1`, `emotion_v2`)

EMOTION_V1

63/63 completed · 18 eligible · 0 flips · 0 findings

0 of 0 cases produced a scored expectation outcome -- no violation rate is available to report (20 excluded, 22 unevaluable, 0 not executed, per the pre-registered oracle rules).

EMOTION_V2

63/63 completed · 18 eligible · 0 flips · 0 findings

0 of 0 cases produced a scored expectation outcome -- no violation rate is available to report (20 excluded, 22 unevaluable, 0 not executed, per the pre-registered oracle rules).

Both endpoints completed the full 63-case seed set (21 baselines + 42 paraphrase/distractor attacks) with zero qualifying flips and zero finding groups on this seed set; a clean result on a small additional suite is evidence about these specific cases, not proof of complete robustness.

sentiment.oces (single target: `sentiment_v1`)

Single-target evaluation of `sentiment_v1` only; comparison is null, matching `sentiment.core`.

60/60 completed · 28 eligible · 2 flips · 2 findings

0 of 0 cases produced a scored expectation outcome -- no violation rate is available to report (12 excluded, 28 unevaluable, 0 not executed, per the pre-registered oracle rules).

OCES.DISTRACTOR/LOGISTICAL: PREDICTION FLIP

1 case(s) in oces.distractor/logistical produced prediction flip -- the top label changed under a meaning-preserving transformation while the clean prediction was confident (at or above 0.6). Severity is scored from the single highest-scoring instance; the 1 affected case(s) indicate reproducibility, not impact.

Observed evidence	1 of 1 cited case row(s) were available; recorded outcomes: HTTP 200: 1; post-request health-check failures: 0.
Why it matters	The top label changed under a transformation intended to preserve meaning, while the clean prediction was confident. This is a model-level sensitivity: the input reached the model unchanged in the ways the input controls check, so no endpoint control addresses it.
Recommended action	This is not fixable by an endpoint control, and adding one would misrepresent the weakness. The appropriate response is a model-level procedure: augment training data with this transformation family and re-evaluate, or document the sensitivity as a known limitation with its measured rate.
How to verify the fix	<pre>python -m runner.run --target-id sentiment_v1 --case-set results/ v63_real_sentiment_oces_087292c1c53c433c/runs/sentiment_oces/ verification_case_set.json --out results/verify_sentiment_v1</pre> after retraining or augmentation, the same case ids are re-run and the flip rate over the same eligible denominator is compared

+ Evidence cases · 1

sst2-val-248.oces.distractor;

OCES.DISTRACTOR/TEMPORAL: PREDICTION FLIP

1 case(s) in oces.distractor/temporal produced prediction flip -- the top label changed under a meaning-preserving transformation while the clean prediction was confident (at or above 0.6). Severity is scored from the single highest-scoring instance; the 1 affected case(s) indicate reproducibility, not impact.

Observed evidence	1 of 1 cited case row(s) were available; recorded outcomes: HTTP 200: 1; post-request health-check failures: 0.
Why it matters	The top label changed under a transformation intended to preserve meaning, while the clean prediction was confident. This is a model-level sensitivity: the input reached the model unchanged in the ways the input controls check, so no endpoint control addresses it.
Recommended action	This is not fixable by an endpoint control, and adding one would misrepresent the weakness. The appropriate response is a model-level procedure: augment training data with this transformation family and re-evaluate, or document the sensitivity as a known limitation with its measured rate.
How to verify the fix	<pre>python -m runner.run --target-id sentiment_v1 --case-set results/ v63_real_sentiment_oces_087292c1c53c433c/runs/sentiment_oces/ verification_case_set.json --out results/verify_sentiment_v1</pre> after retraining or augmentation, the same case ids are re-run and the flip rate over the same eligible denominator is compared

+ Evidence cases · 1

sst2-val-59.oces.distractor;

Evidence-specific remediation

Each finding below follows the same structure: what was observed, why it matters, the recommended action, and the exact command to verify the fix -- reused directly from this run's own recorded remediation_detail evidence where the schema provides it (sentiment.core, sentiment.oces), or the historical finding's remediation text otherwise (emotion.core).

Endpoint / application-layer problems

Critical · malformed/oversized_100kb · unhandled_5xx

MALFORMED/OVERSIZED_100KB: UNHANDLED SERVER ERROR

1 case(s) in malformed/oversized_100kb produced unhandled server error -- the endpoint returned an unhandled 5xx; where health still answered afterwards the request failed but the service survived, which is not unavailability. Severity is scored from the single highest-scoring instance; the 1 affected case(s) indicate reproducibility, not impact.

Recommended action Add a scoped exception handler for this failure mode that returns a typed error response instead of an unhandled traceback.

+ Evidence cases · 1

malformed.oversized_100kb;

High · malformed/oversized_10mb · timeout

MALFORMED/OVERSIZED_10MB: HIT THE REQUEST TIMEOUT

1 case(s) in malformed/oversized_10mb produced hit the request timeout -- no response was produced within the 10 second deadline; the recorded error is client-side and carries no server-stage information, so no cause is attributed. Severity is scored from the single highest-scoring instance; the 1 affected case(s) indicate reproducibility, not impact.

Observed evidence 1 of 1 cited case row(s) were available; recorded outcomes: no response: 1; post-request health-check failures: 0.

Why it matters No response was produced inside the request deadline. The recorded error is client-side and carries no server-stage information, so where the time went is not established by this evidence.

Recommended action Bound the work a single request can do before the deadline: reject oversized bodies at the server layer and cap per-request inference time. Establishing the cause needs server-side timing instrumentation this run did not collect.

How to verify the fix
python -m runner.run --target-id sentiment_v1 --case-set results/v63_real_sentiment_core_bf0c268072bf48dd/runs/sentiment_core/verification_case_set.json --out results/verify_sentiment_v1
the same case returns a bounded response or a documented rejection inside the deadline

+ Evidence cases · 1

malformed.oversized_10mb;

Model-level sensitivity

Medium · encoding/homoglyph · prediction_flip

ENCODING/HOMOGLYPH: PREDICTION FLIP

37 case(s) in encoding/homoglyph produced prediction flip -- the top label changed under a meaning-preserving transformation while the clean prediction was confident (at or above 0.6). Severity is scored from the single highest-scoring instance; the 37 affected case(s) indicate reproducibility, not impact.

Recommended action Model robustness gap rather than an endpoint defect; consider augmenting training data with this perturbation family.

+ Evidence cases · 37

dataset-anger-1.encoding.homoglyph.leading.d3; dataset-anger-2.encoding.homoglyph.leading.d3; dataset-anger-2.encoding.homoglyph.trailing.d3; dataset-anger-2.encoding.homoglyph.trailing.d5; dataset-anger-3.encoding.homoglyph.trailing.d1; dataset-anger-3.encoding.homoglyph.trailing.d3; dataset-anger-3.encoding.homoglyph.trailing.d5; dataset-fear-0.encoding.homoglyph.leading.d5; dataset-fear-0.encoding.homoglyph.middle.d1; dataset-fear-0.encoding.homoglyph.middle.d3; dataset-fear-0.encoding.homoglyph.middle.d5; dataset-fear-0.encoding.homoglyph.trailing.d5; dataset-fear-4.encoding.homoglyph.leading.d5; dataset-joy-0.encoding.homoglyph.leading.d1; dataset-joy-0.encoding.homoglyph.leading.d3; dataset-joy-0.encoding.homoglyph.leading.d5; dataset-joy-3.encoding.homoglyph.trailing.d3; dataset-joy-3.encoding.homoglyph.trailing.d5; dataset-joy-5.encoding.homoglyph.trailing.d5; dataset-sadness-0.encoding.homoglyph.middle.d5; dataset-sadness-3.encoding.homoglyph.leading.d5; dataset-sadness-3.encoding.homoglyph.middle.d5; dataset-sadness-3.encoding.homoglyph.trailing.d5; dataset-sadness-4.encoding.homoglyph.leading.d5; dataset-sadness-5.encoding.homoglyph.leading.d3; dataset-sadness-5.encoding.homoglyph.leading.d5; dataset-surprise-2.encoding.homoglyph.trailing.d1; dataset-surprise-2.encoding.homoglyph.trailing.d3; dataset-surprise-2.encoding.homoglyph.trailing.d5; handwritten-neutral-1.encoding.homoglyph.middle.d3; handwritten-neutral-1.encoding.homoglyph.middle.d5; handwritten-neutral-2.encoding.homoglyph.middle.d5; handwritten-neutral-5.encoding.homoglyph.middle.d1; handwritten-neutral-5.encoding.homoglyph.middle.d3; handwritten-neutral-5.encoding.homoglyph.middle.d5; handwritten-neutral-5.encoding.homoglyph.trailing.d1; handwritten-neutral-5.encoding.homoglyph.trailing.d5;

Medium · perturbation/misspell · prediction_flip

PERTURBATION/MISPELL: PREDICTION FLIP

5 case(s) in perturbation/misspell produced prediction flip -- the top label changed under a meaning-preserving transformation while the clean prediction was confident (at or above 0.6). Severity is scored from the single highest-scoring instance; the 5 affected case(s) indicate reproducibility, not impact.

Observed evidence 5 of 5 cited case row(s) were available; recorded outcomes: HTTP 200: 5; post-request health-check failures: 0.

Why it matters The top label changed under a transformation intended to preserve meaning, while the clean prediction was confident. This is a model-level sensitivity: the input reached the model unchanged in the ways the input controls check, so no endpoint control addresses it.

Recommended action This is not fixable by an endpoint control, and adding one would misrepresent the weakness. The appropriate response is a model-level procedure: augment training data with this transformation family and re-evaluate, or document the sensitivity as a known limitation with its measured rate.

How to verify the fix

```
python -m runner.run --target-id sentiment_v1 --case-set results/v63_real_sentiment_core_bf0c268072bf48dd/runs/sentiment_core/verification_case_set.json --out results/verify_sentiment_v1
```


after retraining or augmentation, the same case ids are re-run and the flip rate over the same eligible denominator is compared

+ Evidence cases · 5

sst2-val-13.perturbation.misspell; sst2-val-211.perturbation.misspell; sst2-val-341.perturbation.misspell; sst2-val-35.perturbation.misspell; sst2-val-52.perturbation.misspell;

Diagnostic observations

Diagnostic cases (pre-registered as having no fixed oracle) are tracked separately from scored findings and never counted toward flip or finding totals. This run's evidence recorded 252 diagnostic observations in the emotion.core historical run; a label change on a diagnostic case is not, by itself, treated as a meaning-preserving vulnerability.

Single-target sentiment observations

High · boundary/length_plus_one · unhandled_5xx

BOUNDARY/LENGTH_PLUS_ONE: UNHANDLED SERVER ERROR

1 case(s) in boundary/length_plus_one produced unhandled server error -- the endpoint returned an unhandled 5xx; where health still answered afterwards the request failed but the service survived, which is not unavailability. Severity is scored from the single highest-scoring instance; the 1 affected case(s) indicate reproducibility, not impact.

Observed evidence	1 of 1 cited case row(s) were available; recorded outcomes: HTTP 500: 1; post-request health-check failures: 0.
Why it matters	Input at or beyond the model's sequence limit produced a server-error response rather than a defined rejection. The settled architecture forbids silent truncation, so rejecting is the intended behaviour and erroring is not.
Recommended action	Check the tokenized length against the model's maximum sequence length before inference and reject over-limit input with a typed 4xx response. Do not add truncation: silently shortening the input would hide the failure this case exists to measure.
How to verify the fix	<pre>python -m runner.run --target-id sentiment_v1 --case-set results/v63_real_sentiment_core_bf0c268072bf48dd/runs/sentiment_core/verification_case_set.json --out results/verify_sentiment_v1</pre> over-limit cases return the documented 4xx rejection and under-limit cases still return 200

+ Evidence cases · 1

boundary.length.plus_one;

TRUNCATION/SIGNAL_START_OVER_LIMIT: UNHANDLED SERVER ERROR

42 case(s) in truncation/signal_start_over_limit produced unhandled server error -- the endpoint returned an unhandled 5xx; where health still answered afterwards the request failed but the service survived, which is not unavailability. Severity is scored from the single highest-scoring instance; the 42 affected case(s) indicate reproducibility, not impact.

Observed evidence	42 of 42 cited case row(s) were available; recorded outcomes: HTTP 500: 42; post-request health-check failures: 0.
Why it matters	Input at or beyond the model's sequence limit produced a server-error response rather than a defined rejection. The settled architecture forbids silent truncation, so rejecting is the intended behaviour and erroring is not.
Recommended action	Check the tokenized length against the model's maximum sequence length before inference and reject over-limit input with a typed 4xx response. Do not add truncation: silently shortening the input would hide the failure this case exists to measure.
How to verify the fix	<pre>python -m runner.run --target-id sentiment_v1 --case-set results/ v63_real_sentiment_core_bf0c268072bf48dd/runs/sentiment_core/ verification_case_set.json --out results/verify_sentiment_v1</pre> over-limit cases return the documented 4xx rejection and under-limit cases still return 200

+ Evidence cases · 42

```
sst2-val-109.truncation.signal_start_over_limit; sst2-val-110.truncation.signal_start_over_limit; sst2-val-111.truncation.signal_start_over_limit; sst2-
val-120.truncation.signal_start_over_limit; sst2-val-121.truncation.signal_start_over_limit; sst2-val-13.truncation.signal_start_over_limit; sst2-
val-136.truncation.signal_start_over_limit; sst2-val-162.truncation.signal_start_over_limit; sst2-val-17.truncation.signal_start_over_limit; sst2-
val-18.truncation.signal_start_over_limit; sst2-val-201.truncation.signal_start_over_limit; sst2-val-211.truncation.signal_start_over_limit; sst2-
val-227.truncation.signal_start_over_limit; sst2-val-248.truncation.signal_start_over_limit; sst2-val-290.truncation.signal_start_over_limit; sst2-
val-319.truncation.signal_start_over_limit; sst2-val-341.truncation.signal_start_over_limit; sst2-val-35.truncation.signal_start_over_limit; sst2-
val-361.truncation.signal_start_over_limit; sst2-val-427.truncation.signal_start_over_limit; sst2-val-432.truncation.signal_start_over_limit; sst2-
val-480.truncation.signal_start_over_limit; sst2-val-489.truncation.signal_start_over_limit; sst2-val-52.truncation.signal_start_over_limit; sst2-
val-561.truncation.signal_start_over_limit; sst2-val-588.truncation.signal_start_over_limit; sst2-val-59.truncation.signal_start_over_limit; sst2-
val-590.truncation.signal_start_over_limit; sst2-val-601.truncation.signal_start_over_limit; sst2-val-609.truncation.signal_start_over_limit; sst2-
val-656.truncation.signal_start_over_limit; sst2-val-671.truncation.signal_start_over_limit; sst2-val-672.truncation.signal_start_over_limit; sst2-
val-674.truncation.signal_start_over_limit; sst2-val-686.truncation.signal_start_over_limit; sst2-val-730.truncation.signal_start_over_limit; sst2-
val-743.truncation.signal_start_over_limit; sst2-val-748.truncation.signal_start_over_limit; sst2-val-761.truncation.signal_start_over_limit; sst2-
val-778.truncation.signal_start_over_limit; sst2-val-785.truncation.signal_start_over_limit; sst2-val-805.truncation.signal_start_over_limit;
```

OCES.DISTRACTOR/LOGISTICAL: PREDICTION FLIP

1 case(s) in oces.distractor/logistical produced prediction flip -- the top label changed under a meaning-preserving transformation while the clean prediction was confident (at or above 0.6). Severity is scored from the single highest-scoring instance; the 1 affected case(s) indicate reproducibility, not impact.

Observed evidence	1 of 1 cited case row(s) were available; recorded outcomes: HTTP 200: 1; post-request health-check failures: 0.
Why it matters	The top label changed under a transformation intended to preserve meaning, while the clean prediction was confident. This is a model-level sensitivity: the input reached the model unchanged in the ways the input controls check, so no endpoint control addresses it.
Recommended action	This is not fixable by an endpoint control, and adding one would misrepresent the weakness. The appropriate response is a model-level procedure: augment training data with this transformation family and re-evaluate, or document the sensitivity as a known limitation with its measured rate.
How to verify the fix	<pre>python -m runner.run --target-id sentiment_v1 --case-set results/ v63_real_sentiment_oces_087292c1c53c433c/runs/sentiment_oces/ verification_case_set.json --out results/verify_sentiment_v1</pre> after retraining or augmentation, the same case ids are re-run and the flip rate over the same eligible denominator is compared

+ Evidence cases · 1

sst2-val-248.oces.distractor;

OCES.DISTRACTOR/TEMPORAL: PREDICTION FLIP

1 case(s) in oces.distractor/temporal produced prediction flip -- the top label changed under a meaning-preserving transformation while the clean prediction was confident (at or above 0.6). Severity is scored from the single highest-scoring instance; the 1 affected case(s) indicate reproducibility, not impact.

Observed evidence	1 of 1 cited case row(s) were available; recorded outcomes: HTTP 200: 1; post-request health-check failures: 0.
Why it matters	The top label changed under a transformation intended to preserve meaning, while the clean prediction was confident. This is a model-level sensitivity: the input reached the model unchanged in the ways the input controls check, so no endpoint control addresses it.
Recommended action	This is not fixable by an endpoint control, and adding one would misrepresent the weakness. The appropriate response is a model-level procedure: augment training data with this transformation family and re-evaluate, or document the sensitivity as a known limitation with its measured rate.
How to verify the fix	<pre>python -m runner.run --target-id sentiment_v1 --case-set results/ v63_real_sentiment_oces_087292c1c53c433c/runs/sentiment_oces/ verification_case_set.json --out results/verify_sentiment_v1</pre> after retraining or augmentation, the same case ids are re-run and the flip rate over the same eligible denominator is compared

+ Evidence cases · 1

sst2-val-59.oces.distractor;

ci.emotion release-gate evidence

The release gate's direct users are developers and maintainers deciding whether a candidate build may deploy; everyday users of a deployed model benefit only indirectly, because a weak candidate is blocked before it ships. The gate currently tests one candidate: hardened Emotion V2, evaluated as ci.emotion, against a fixed 162-case selection of the core suite and a frozen policy.

Candidate emotion_v2 · **Reference** ci_core_v1_clean_reference · **Policy** ci_emotion_v2_core_selection v1.0.0

Outcome pass · **Exit code** 0 · 13/13 checks passed

Exit 0 = pass (deployment may proceed); exit 1 = policy failure (a check did not meet the frozen policy); exit 2 = execution error (the run itself could not be completed, e.g. an unreadable policy or an endpoint that never started). Both failure types block the workflow's deploy job.

CHECK	STATUS
policy_valid	pass
candidate_identity	pass
selection_integrity	pass
selection_completion	pass
clean_responses_coherent	pass
no_server_errors	pass
no_timeouts_or_transport_failures	pass
health_after_every_request	pass
no_leak_signatures	pass
documented_rejection_behaviour	pass
clean_label_agreement	pass
ground_truth_baseline_accuracy	pass
slow_responses	pass

The authoritative CI execution for this line of commits is GitHub Actions run 35075844725 ('Stage 3 release gate', PR #11, stage3/ahsan), confirmed via the public GitHub REST API to have completed with conclusion success and its deploy job skipped. This session has no GitHub API token to download that run's own uploaded artifact bytes, so it is cited by run number and verified outcome only, never hashed as if its artifact

were present. The gate evidence rendered above (outcome, exit code, all 13 checks) is instead a fresh, real local reproduction of the identical command at this report's own application commit -- not a copy of that run's artifact -- see [artifacts/ci_gate_evidence_v1/PROVENANCE.md](#).

This is a working reference implementation the gate exercises against one approved emotion release candidate; it does not universally certify arbitrary models. A developer adapting the toolkit to a new target must register that target, author its own case selection, and freeze its own policy.

What this report is built from

Evidence-generation Application Commit

78188adbdc4449429116af3161a032516d38aa8

Emotion Model / Tokenizer

j-hartmann/emotion-english-distilroberta-base @ 0e1cd914e3d4...

Sentiment Model / Tokenizer

distilbert/distilbert-base-uncased-finetuned-sst-2-english @ 714eb0fa89d2...

Contracts Version

3.0.0

CI Policy Identity

ci_emotion_v2_core_selection v1.0.0 (sha256 e4cb0a86a3590037...)

Case-set Identity

ci_core_v1

OCES Generator Version

1.2.1

Report-generation Identity

report.master_report v1.0.0

The evidence above (sentiment.core, emotion.oces, sentiment.oces, the local ci.emotion gate reproduction) was generated by running the real pipeline against live endpoints at application commit 78188adbdc4449429116af3161a032516d38aa8. This HTML page was rendered afterward, from that already-recorded evidence, by report.master_report; report generation reads and formats evidence, it never re-scores or re-runs it.

Evidence file hashes

Every quantitative claim above is derived from these files, re-hashed and checked against report/master_evidence/manifest.json at generation time. A missing or modified file fails generation closed (no report is produced).

EVIDENCE	PATH	SHA-256	EVALUATION
Historical V5 full emotion Core benchmark analysis (V1+V2 paired, 1928/1928 each).	artifacts/verified_full_report/analysis.json	12d47b35c700c6c1...	emotion.core
Export hash metadata for the archived /verified-full/ bundle.	artifacts/verified_full_report/export_meta.json	6e445d037ce485d9...	emotion.core
Archived V5 rendered report, served at /verified-full/.	artifacts/verified_full_report/report.html	ab66edbfec500e7e...	emotion.core

EVIDENCE	PATH	SHA-256	EVALUATION
Byte-provenance and reproduction proof for the emotion.core historical run.	artifacts/benchmark_v6/ PROVENANCE.md	151bddcf1f11dab4...	emotion.core
Frozen request manifest for the emotion.core historical run (both endpoints).	artifacts/benchmark_v6/ run/manifest.json	0471bccbf293095d...	emotion.core
Real single-target sentiment.core run analysis (schema v3).	artifacts/ sentiment_core_v1/ analysis/analysis.json	7847d2c4bbd6cca7...	sentiment.core
Provenance and verified counts for the sentiment.core run.	artifacts/ sentiment_core_v1/ PROVENANCE.md	5ac00da6535b08d2...	sentiment.core
Real paired emotion.oces run analysis (schema v3).	artifacts/ emotion_oces_v1/ analysis/analysis.json	cb6d9c3531d60b05...	emotion.oces
Provenance and verified counts for the emotion.oces run.	artifacts/ emotion_oces_v1/ PROVENANCE.md	2da16725be7c0934...	emotion.oces
Frozen-defense content hashes recorded at emotion.oces run time.	artifacts/ emotion_oces_v1/run/ oces_freeze_content_hashes.json	f39fd296a49e3f02...	emotion.oces
Real single-target sentiment.oces run analysis (schema v3).	artifacts/ sentiment_oces_v1/ analysis/analysis.json	18bc903eb63367bb...	sentiment.oces
Provenance and verified counts for the sentiment.oces run.	artifacts/ sentiment_oces_v1/ PROVENANCE.md	f83cf273bc375d1a...	sentiment.oces
Pre-registered OCES case authoring sheet: 82 rows, proposed_tier column (0 GOLD / 50 SILVER / 32 REVIEW).	attacks/data/oces/ oces_review_sheet.csv	59791c1332c1ea51...	—
Real local ci.emotion gate outcome (pass, exit 0, 13/13 checks) at this milestone's commit.	artifacts/ ci_gate_evidence_v1/ gate_result.json	e20f158da1d5b914...	ci.emotion
Analysis document backing the local ci.emotion gate run.	artifacts/ ci_gate_evidence_v1/ analysis/analysis.json	af9d5357a4aa63a3...	ci.emotion

EVIDENCE	PATH	SHA-256	EVALUATION
Provenance for the local ci.emotion gate reproduction, incl. its relationship to the cited GitHub Actions run.	artifacts/ ci_gate_evidence_v1/ PROVENANCE.md	b4c6ab200b843410...	ci.emotion
Frozen CI release-gate policy (Khalid-owned, analysis/policies/).	analysis/policies/ ci_core_v1.json	e4cb0a86a3590037...	ci.emotion
Static, model-free target/evaluation registry defining all five evaluation identities.	endpoint/targets.json	3212f30cde643ba6...	—

External citations (not independently hashed)

GitHub Actions 'Stage 3 release gate' run on stage3/ahsan (PR #11), confirmed via the public GitHub REST API to have completed with conclusion success and its deploy job skipped. This session has no GitHub API token to download the run's own uploaded artifact bytes, so it is cited by run number and verified outcome only, never hashed as if its artifact were present. Run 35075844725.

Explicitly not applicable

- **sentiment.core / sentiment.oces hardening comparison** — There is no sentiment_v2 target. comparison is null in both analysis documents by contract; no before/after claim is made.
- **emotion.core / emotion.oces model revision independent attestation** — artifacts/benchmark_v6/PROVENANCE.md: the historical run predates revision pinning; the revision is strongly supported (re-verified cache scan) but not recorded by the run itself.
- **OCES GOLD-tier cases** — attacks/data/oces/oces_review_sheet.csv proposed_tier column contains zero GOLD rows (0 GOLD / 50 SILVER / 32 REVIEW of 82).

Limitations

- Bounded to English text classification; the toolkit does not evaluate arbitrary modalities or languages.
- There is no arbitrary public-model-upload or external-endpoint-URL feature; a new target must be registered in endpoint/targets.json by someone with repository access.
- Sentiment has no V2 and therefore no hardening-improvement comparison; comparison is null everywhere sentiment appears.
- OCES is authored by the team after the evaluated defenses were frozen -- it is not blind or externally authored, and is not a substitute for an independent audit.
- OCES contains zero GOLD-tier cases across both emotion and sentiment (0 of 82); most of its evidentiary weight is SILVER, with a real REVIEW-tier minority.
- The CI release gate (ci.emotion) currently certifies only the approved emotion_v2 candidate against its own frozen policy; it does not certify sentiment or any other target.
- A robustness evaluation of named adversarial inputs does not prove complete safety, fairness or general correctness of the underlying model.
- Diagnostic truncation cases are pre-registered as having no fixed oracle and are never automatically treated as evidence of a meaning-preserving vulnerability, even when the observed label changes.

Historical Red Lab v5 benchmark

The Red Lab v5 emotion benchmark (the 1928-case, 218/112-flip V1/V2 comparison presented in Section D) remains archived at /verified-full/ exactly as originally published, with its own V5 branding and hashes unchanged by this milestone. It is linked here as historical supporting evidence, not re-rendered or re-scored.

Open the archived Red Lab v5 report · [Source JSON](#)